Course milestone M9: Bounded Research Note and Claim-Evidence Table

HONR 46400 · Evidence-Driven Research

Start here

Your main file is `lastname_m09.pdf`. Build it in this order:

1. **Your carry-forward record, at the very top.** Open my review email and your research action plan side by side. Answer every numbered request for this milestone (each has an ID like `IR-M09-01`) and every step your plan sets for this milestone. Next to each one, write **Done** and exactly where I can find it, or **Not adopted**, with your reason and what you chose instead. Several items can point to the same piece of work.
2. **Then your milestone work and files.** The book milestone below explains the research work. The complete course checklist, the files to hand in and the rubric are in this milestone's assignment on Brightspace: check them before you submit.
3. **The Expo work** in *What this course adds*, below.

How this counts. Your answer to the action plan is worth 10 points of the rubric, scored for you individually. Each numbered request left unanswered costs 5 points, up to 20. Only requests and plan steps I sent you at least 48 hours before the deadline count; anything later moves to the next milestone. A step that is also a numbered request is counted once, as a request. If I sent you no numbered requests, or no plan steps apply, say so in one line. Not sure what applies to you? Write to me before the deadline; that is exactly what I am here for.

Submit it on Brightspace, under **Assignments** then **M09**. Brightspace carries the deadline.

Your work lives in a Colab notebook, which has no export-to-PDF button. **Making the PDF you hand in**, at the end of this document, is the one-minute routine that turns the notebook into the file you upload.

This milestone presents **Book Milestone 9: Your bounded claims** (checkpoint, version 1) of the course book, EDR|AI.

What this course adds

Your note is your poster's first full draft

The book page below asks for a stand-alone research note. This course fixes its shape, because your next artifact is a printed poster at the undergraduate research expo: you write the note under the sections of the course poster template (`honr464_poster_template.pptx`, 34 × 22.5 inches), in the template's order. When the poster draft is due, every panel already has its text, written in full sentences and traced.

Open with your working title and your **headline sentence**: the bounded claim, with its interval or its boundary in the same sentence, as it will run in the poster's headline banner. Then write the thirteen sections of the course poster template, in its order, each under its own heading:

#	Section	What your note writes there
1	**Abstract**	Your conference abstract, updated to your verified results: one paragraph of up to 150 words with the question and its answer, what the literature concludes and what you add, the data and the design, and the main result with its uncertainty and the limitation that matters most. Add one line saying what changed since you submitted it.
2	**The question and why it matters**	Your research question in one sentence, as your Research Contract declares it, with its kind and its reach; why it matters, and to whom; your main hypothesis as you recorded it before the analysis ran.

#	Section	What your note writes there
3	**What is already known**	The main conclusion of the literature, from at least two sources you read yourself; the gap; and what your project adds. With the question above, this is your problem-gap-question spine.
4	**Design: what your claim needs**	Kind, reach, target and warrant, one line each, plus the identification status. A causal question your design cannot identify is a causal target, currently unidentified, never relabeled descriptive.
5	**Data and measurement**	Source and route with retrieval dates; observations kept apart from independent units; the time window; each concept and the measure that stands for it; missing data; permissions.
6	**Methods**	The analysis you declared before seeing results and why it fits your design, with any later change and its reason; how you computed the uncertainty, at the level of your independent unit, with any dependence named; software and seed.
7	**Main result**	The result in numbers and units, with its uncertainty in the same sentence: the interval or range your design supports or, where it genuinely carries none, the precise reason why. Then the figure you plan to print, captioned with the notebook cell that drew it.
8	**Checks and verification**	The checks from your robustness audit and what each one changed, and what has been reproduced so far, by whom and when.
9	**Conclusions**	Your bounded headline claim, carrying its kind and its reach, then what this evidence supports and what it does not yet support.
10	**Limitations**	The two or three limitations that genuinely threaten the claim, led by the one your robustness audit could not fix, each with the next step that would settle it.
11	**AI use disclosure**	Drafted from your AI Research Ledger: for each tool, the task, what you changed or kept, and how you checked it; what you sent to tools; and what you did not verify.
12	**References**	Every source the note cites, each one retrievable: author, year, title, venue, DOI or URL.
13	**Acknowledgments**	One line is enough: your mentor, and anyone who provided data, took part, or funded the work.

Write every section in full sentences, in two to four pages, not counting the references and the claim-to-evidence table; *Acknowledgments* may be one line. The book's practice lives inside the sections: the claim with its boundary in your headline sentence and *Conclusions*, uncertainty and limitations in *Main result* and *Limitations*, and the disclosure from your ledger in *AI use disclosure*. The claim-to-evidence table stays its own section of your file.

Your claim-to-evidence table travels to the poster

Mark every row with the milestone that added it, this milestone for every row now, and keep the table after you submit it. The poster may grow beyond the note: a result, a figure, a check, or a source that improves your work later is welcome on the board. It enters the way everything in the note did, as a new row in the table, traced to the notebook cell that computed it or the exact place in its source, before it goes on the poster. You hand in the table's current version again with the poster draft and with the locked poster.

Milestone 9: Your bounded claims

Open the milestone workbook in Colab

This chapter closes Studio 9: Write, bound, and disclose. Its lessons each ended at **It is your turn**; what you wrote there arrives here as working material, and this is where the pieces become one artifact you can defend.

What this milestone produces. A stand-alone research note or report draft, a claim-to-evidence table, and your AI disclosure statement.

What you bring

You also carry forward the last milestone's artifact: **Milestone 8: Your robustness audit**. This one builds on it, and the chain assumes it is versioned and filed.

Check you are carrying each of these before you start. If one is missing, go back and work that lesson's **It is your turn** first; the practice below assumes it.

- Lesson 28: From Results to Claims: your claims with verbs downgraded to what the design established, each carrying its boundary line.
- Lesson 29: Claim–Evidence Tables: your claim-evidence table, one row per claim with claim, evidence, trace, and boundary, every number traced to its output.
- Lesson 30: AI Disclosure and Research Integrity: your AI-use disclosure, written from the ledger and naming what you did not verify.
- Lesson 31: From Dossier to Research Note: your stand-alone note draft: the problem-gap-question spine, methods, results with uncertainty, discussion, ordered findings with the headline claim's boundary inside, references, your disclosure placed, and the pointer to the package Studio 11 will freeze.

The practice

1. Write your claim as one sentence, with its boundary attached in the same sentence.
2. Build the claim-to-evidence table: every claim, the output that supports it, and the check that verified it.
3. State your uncertainty and your limitations in the words a reader outside your field would use.
4. Write your AI disclosure from your ledger, not from memory.

A version, not a pass

Your milestone artifact is a dated, numbered **version** with the reason for the version attached. When later evidence changes it, you write the next version rather than editing the last one, because the sequence of changes is itself part of your research record.

The four rails, here

Four concerns cross every studio in this book. At this milestone they take these specific forms.

- **Ethics, permissions, and data exposure.** Disclosure includes what you sent to tools, not only what they produced.
- **Evidence, provenance, and reproducibility.** A claim with no row in the table does not appear in the draft.
- **AI activity, verification, and human decisions.** The sentence that ships is yours, whatever drafted it.
- **Uncertainty, claim boundary, and revision history.** Findings are never communicated as certainties.

Where this milestone sits

You arrived carrying Milestone 8: Your robustness audit. Milestone 10 adapts these bounded claims to a real audience and rehearses their defense. The chain ends at Milestone 12: Your release and next cycle, where you decide whether your finished research artifact leaves your hands.

What sends you back

A claim you cannot fit in the table is a claim to cut or a check to run, and both send you back a studio.

The book's own checks

Your course rubric on Brightspace scores these checks inside its rows, so use this table to test your work before you hand it in. The points below are the book's, not your course score.

Each row scores **0**, **1**, or **2**. **10 points total.**

#	Criterion	0	1	2
1	The milestone artifact exists in full, specific to your own project	absent, or generic text that would fit any project	present but incomplete, or specific in some parts and generic in others	complete and unmistakably about your project

#	Criterion	0	1	2
2	The milestone is written as a dated, numbered version with its reason	no version, or a version with no reason given	versioned, but the reason is decorative rather than an account of what changed	versioned with a reason a reader could use to reconstruct your thinking
3	Every judgment in the artifact is defended by you, not asserted by a tool	conclusions appear with no reasoning, or reproduce a tool's wording	reasoning present but thin where the decision was hardest	each judgment carries a reason you could defend under questioning
4	The four rails are carried in the form this studio requires	one or more rails absent where this studio required them	all rails present but one is nominal	every rail addressed in the form this studio requires
5	The milestone's own product is present and defensible: A stand-alone research note or report draft, a claim-to-evidence table, and your AI disclosure statement	the milestone's own product is missing	present but would not survive a careful question	present and defensible as stated

Your workbook

Open the workbook with the link above. It walks these steps with a cell for each one, ends by writing your milestone version with its reason, and adds the rows this studio contributes to your AI Research Ledger and your Research Dossier.

Making the PDF you hand in

Colab has no export-to-PDF button. You make the PDF by **printing the notebook to a file**, which takes about a minute once you know the preparation steps. Skipping them is the most common way a milestone arrives looking half finished, because a notebook prints what is on the screen and nothing else.

1. Prepare the notebook

1. **Run the whole thing.** `Runtime` → `Run all`, then wait for it to finish. A cell you never ran prints with nothing underneath it, so an analysis you really did can reach the grader as an empty box.
2. **Expand every collapsed section.** Anything folded shut prints as a heading with nothing beneath it. Open all of them, including any Colab folded for you.
3. **Shorten any output that scrolls.** When Colab puts a scrollbar on an output, or tells you the output is truncated, only the visible part prints. Show the first twenty rows instead of the whole table, or summarise it, then rerun that cell. A reader did not want the whole table anyway.
4. **Read it once, top to bottom.** You are about to freeze it.

2. Print it to a file

`File` → `Print` opens your browser's print dialog. Set the destination to **Save as PDF**, open the dialog's further settings, and turn on **Background graphics**, so code cells keep their shading and your figures keep their backgrounds. Save it under the file name this milestone asks for.

Wide tables and wide figures are the usual casualty. If something runs off the right edge, switch the layout to **landscape** or reduce the scale, and print again.

3. Check the PDF before you upload it

Open the file you just made and confirm three things: every figure is whole rather than sliced by a page break, no output stops mid-line, and the pages run in order with nothing missing. A PDF is an artifact like any other in this course, so you verify it before your name goes on it.

One route not to take

Ask an AI assistant how to export a Colab notebook and it will very likely hand you `jupyter nbconvert --to pdf`. Do not spend your afternoon on it. Inside Colab that route needs a large LaTeX installation, runs for several minutes, and then either fails or quietly drops the symbols and emoji these notebooks are full of. Printing to a file is the route that works.

That is a small, cheap instance of the rule this whole course runs on. The confident answer is not automatically the correct one, and the way you find out is to check it against what actually happens.